

Antoine Boucher

SOFTWARE ENGINEER — PLATFORM AND COMPUTER GRAPHICS

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Profile

Software engineer focused on cloud-native backend systems, DevSecOps automation, and developer tooling. Strong experience with Java/Kotlin and Python, Docker/Kubernetes, CI/CD, and infrastructure-as-code. Master's in Information Technology Engineering from ÉTS.

Core Skills

Backend	Java, Kotlin, Python, C++, REST APIs, GraphQL, Microservices, OpenGL
DevOps / Cloud	Docker, Kubernetes, Terraform, GitLab CI, GitHub Actions, Nginx, GCP, Azure
Data / Storage	PostgreSQL, MySQL, MongoDB, Redis, Kafka, Data Science, Business Research
Testing / Quality	JUnit, Integration/E2E Testing (Playwright), Playwright, Observability/Logging
Other	Linux, Bash, Ansible, Security-minded delivery
AI coding / Agents	Claude, MCP, AI-assisted coding, software agents

Experience

IMC2 – Institut multidisciplinaire en cybersécurité et cyberrésilience

Remote (Montréal, QC)

Research Assistant (Permanent Part-time)

Mar 2025 – Feb 2026

- Managed CI/CD pipelines with GitLab CI and GitLab Runners for end-to-end DevSecOps workflows, improving deployment reliability
- Developed Java microservices with robust unit tests (JUnit) and integration/E2E tests (Playwright), achieving high test coverage
- Containerized services with Docker and deployed with Kubernetes, Terraform, and Nginx on GCP, streamlining infrastructure management
- Created a cybersecurity risk-scoring algorithm and an interactive risk-visualization UI (Angular) for enhanced threat analysis
- Technical Skills:** Java, Kotlin, Docker, Kubernetes, Terraform, GitLab CI, GCP, Nginx, JUnit, Playwright, Angular

Polytechnique Montréal

Montréal, QC (On-site)

Teaching Assistant, LOG8100 DevSecOps (Contract Part-time)

Sep 2024 – Jan 2026

- Created and improved course/lab content focused on secure software delivery (DevSecOps)
- Set up and maintained the GitLab project structure for 20 teams (repositories, CI/CD templates, runners/workflows), and managed permissions/access control
- Assisted teams with Docker and Kubernetes fundamentals (containerization, deployments, debugging)
- Supported security learning with practical references to OWASP (common web risks, secure practices, and remediation mindset)
- Corrected practical assignments and provided technical coaching to unblock teams and improve deliverables
- Technical Skills:** GitLab, CI/CD, DevSecOps, Docker, Kubernetes, OWASP, Secure Coding, Linux

École de technologie supérieure (ÉTS)

Montréal, QC (Hybrid)

Teaching Assistant (Contract Part-time)

Feb 2024 – Apr 2026

- Courses: TCH057 (Winter 2024), IND500 (Fall 2025), TCH099 (Summer 2024, Winter 2025, Summer 2025), GTI660 (Summer 2024), GTI320 (Winter 2026)
- Built and refreshed lab material for distributed data topics (SQL, MongoDB, Kafka, PostgreSQL); designed assignments, rubrics, and detailed feedback for IND500, GTI660, and related cohorts
- Supported integrator projects (TCH099): debugging, architecture, backend and data pipelines; helped maintain course infrastructure including Microsoft Azure; coached teams on delivery, milestones, and SCRUM basics
- Supported labs for mobile/UX (TCH057) and course delivery for GTI320 (Winter 2026)
- Technical Skills:** SQL, MongoDB, Kafka, PostgreSQL, Distributed Databases, Microsoft Azure, Android Studio, UI/UX, SCRUM

IONODES

Laval, QC (Hybrid)

Cloud Developer (Permanent Full-time)

May 2023 – Aug 2023

- Designed and implemented a 3-tier subscription system with organization-based permissions using Auth0 for the IoT platform
- Developed Sentry SDK integrations in C# to enhance logging with Serilog, improving error tracking and monitoring
- Acquired expertise in ONVIF and WebRTC protocols for IoT video streaming applications
- Orchestrated deployments through Azure DevOps with feature flags and structured release workflows
- Technical Skills:** C#, .NET, Azure DevOps, Auth0, Sentry, Serilog, UML, Microservices, ONVIF, WebRTC

Intact

Back End Developer (Co-op)

Montréal, QC (Hybrid)

Jan 2022 – May 2022

- Created new campaigns based on client sign-up dates, providing personalized experiences for new clients
- Implemented API endpoint logic for enrolling new clients on legacy systems, streamlining the onboarding process
- Developed backend login and MFA API for mobile apps using Kotlin, providing secure and easy access for clients
- Improved ELK stack by creating Kibana dashboards for effective error management and defined error codes across multiple microservices
- **Technical Skills:** Kotlin, Spring Boot, GraphQL, MongoDB, Microservices, Docker Swarm, Jira, Kibana, SAFe

Divine Essence (Union-Nature)

Web Specialist (Contract Part-time)

Dorval, QC (On-site)

Jun 2020 – Dec 2020

- Redesigned bilingual blog (French and English), improving user experience and accessibility
- Implemented custom dynamic sales labels for search optimization on CommerceBuild, enhancing SEO performance
- Built custom "Add to Cart" API call workflow, streamlining e-commerce transactions
- Provided technical support and internal training on HTML, CSS, and JavaScript fundamentals
- Fixed popup, banner, and grid CSS issues; deployed a 3CX chatbot integration for customer support
- **Technical Skills:** HTML, CSS, JavaScript, CommerceBuild, 3CX, SEO

Power Go

Full Stack Web Developer (Co-op)

Montréal, QC

Sep 2020 – Apr 2021

- Utilized Python, Django, Vue.js, and PostgreSQL to develop a versatile data parser capable of handling XML, JSON, CSV, and Excel formats
- Integrated APIs for Facebook Shop and Marketplace, expanding sales channels
- Implemented a currency exchange rate system for CAD/USD conversions for complete inventory system and date synchronization
- **Technical Skills:** Python, Django, Vue.js, PostgreSQL, MongoDB, JavaScript, REST API, Git, Jira, Full-Stack Development

Wandrian

Full Stack Developer (Co-op → Part-time)

Montréal, QC

May 2019 – Jan 2020

- Travel/ticketing (Italiarail): worked with Railkey Tech on performance and UX; Braintree payment integration; UI redesign for consistency; ticketing frontend refresh to new guidelines; purchase-related API endpoints and bugfixes
- Internal tooling: Django, Pyramid, and jQuery for customer-service web tools and a currency-usage analysis dashboard
- Quality: rewrote Selenium end-to-end test infrastructure during co-op and extended automated coverage while part-time
- **Technical Skills:** Python, Django, Pyramid, JavaScript, jQuery, Selenium, MongoDB, REST API, Git, Jira, Test Automation

Algolux

Engineering Intern

Montréal, QC (On-site)

May 2017 – Aug 2017; May 2018 – Aug

2018

- Automation for camera/ISP validation: Ansible and Linux/QEMU UEFI VM workflows for REST-based installs; HDR 10bpp test imagery for Nexus 6 ISP; Flask REST services for remote display and capture workflows
- Vision data and analysis: large-scale outdoor capture and annotation pipeline (~12k images); Python tool for raw color analysis; software control for camera rigs/tripods for varied viewpoints
- Lab operations: QNAP backups for Ubuntu workstations, networking support, Linux/Windows desktops and servers for imaging experiments and REST tooling
- **Technical Skills:** Python, C++, CUDA, OpenCV, Flask, Ansible, Linux, Windows, REST APIs

École de technologie supérieure (ÉTS)

Manager, digital fabrication lab (Part-time)

Montréal, QC (On-site)

Sep 2016 – Apr 2023

- Operated electronics and software development lab (3D printers/VR/electronics)
- Ensured materials and tools were always available for users; assisted students with 3D printers and VR equipment
- Helped troubleshoot technical issues and guided students in innovative projects; maintained and repaired 3D printers
- **Technical Skills:** 3D Printing, Electronics, Equipment Maintenance

Collégial International Sainte-Anne

Manager of the MakerSpace (Part-time)

Montréal, QC (On-site)

Jan 2016 – May 2017

- Ran the MakerSpace (3D printers/VR/electronics), maintained equipment, and supported student projects end-to-end
- Assisted students with digital fabrication, prototyping, and innovation projects
- Ensured materials and tools were always available for users; promoted innovation and creativity in digital fabrication and virtual reality
- **Technical Skills:** 3D Printing, VR, Electronics, Equipment Maintenance, Digital Fabrication

Leadership & Community

- Maintains club infrastructure and web presence; supports projects in trading, data, and engineering education
- Leads development and support for internal tools (platform, strategies, workshops, automation)
- **Technical Skills:** Python, Trading Platforms, Data Science, Backtesting, Infrastructure

Education

École de technologie supérieure (ÉTS) Montréal, QC, Canada
Master’s Degree, Information Technology Engineering Aug 2023 – Apr 2026

- Thesis research: real-time surface wear in interactive physics simulations; dynamic friction and texture.

École de technologie supérieure (ÉTS) Montréal, QC, Canada
Bachelor’s Degree, Information Technology Engineering Sep 2018 – Apr 2023

Certifications & Programs

Certification	AWS Certified Cloud Practitioner
Programs	Programme Québécois d’Entrepreneuriat Scientifique (QcES)
Cloud Quest	Data Analytics, Networking, Cloud Practitioner, Machine Learning, Serverless Developer, Solutions Architect
Other training	Udemy: Ultimate AWS Certified Cloud Practitioner CLF-C02 (Mar 2024)
Conference Presentations	Presenter, Graphquon 2025 (Nov 15–16, 2025) — Annual Quebec-Ontario Pre-SIGGRAPH Workshop, University of Toronto

Selected Projects

MarkÉTS & AlgoÉTS Trading Platform ÉTS Algorithmic Trading Club
Fintech / Algorithmic Trading Platform Jan 2023 – Apr 2026

- Developed and maintained MarkÉTS paper-trading competition platform (170+ participants, 4,154 orders, 99%+ uptime over a 4-month competition) and core trading infrastructure (orders, portfolios, leaderboards)
- Maintained club website (algoets.etsmtl.ca) with member management, educational content, and platform integration; built Python/Pandas strategies and backtesting workflows
- **Technical Skills:** Python, Django, PostgreSQL, Real-time Systems, Financial APIs, Backtesting, Portfolio Management, Infrastructure

Personal Homelab & Multi-Cloud Practice Self-hosted
Infrastructure / Homelab

- Built and operate a Proxmox-based homelab for virtualization, networking, and repeatable lab environments
- Hands-on deployment and monitoring patterns across multiple public clouds for distributed workloads and observability
- **Technical Skills:** Proxmox, virtualization, multi-cloud deployment, monitoring and observability

uml-mcp Open Source
Developer Tools / AI Coding Assistants

- MCP (Model Context Protocol) server exposing UML/diagram tools to LLM-based coding assistants, to produce architecture from natural language or repository context and automate technical documentation
- **Technical Skills:** MCP, LLM tool integration, technical documentation automation